

Status_Summary

RK3588 / Orange Pi 5 Max — A/B-virt Emulator — Status Summary

Revision: 2 **Last modified:** 2026-06-11T12:30:00Z **Companion of:** [Status.md](#) (§11.4.56 two-audience parity).

Page 1 — For the operator (plain language)

We are building a way to test the RK3588 / Orange Pi 5 Max over-the-air (OTA) update flow **without needing the actual board on the desk yet** — by emulating the device on this Mac.

What works today:

- The emulated device **boots up and runs**. We built a small aarch64 Linux image and it boots all the way to a working login on this Mac (using Apple's built-in virtualization). We have the full boot recording as proof.
- The container-based test (control plane talking to a fake device) also works end-to-end.
- Our automated quality bank for this project passes its checks (12 of 12), including the check that the boot recording above is real and on file.

What is still pending (NOT done yet — honestly):

- The real “switch to the new system half and roll back if it fails” part is **NOT proven yet**. That is the heart of the A/B update (pick slot A or B, verify it, fall back if the new one is bad). The scripts that do this — the two-slot disk builder and the bootloader slot-picker / auto-rollback logic — are now **written and check out cleanly, but have not been run**: they need a real bootloader binary that an image build (running right now) will produce. Until we actually run them and capture the result, we do NOT claim it works.
- The full real-Android A/B update test needs a **Linux machine** (it cannot run on this Mac — the Mac lacks the required virtualization device). It is ready to run the moment your incoming Linux host is available; for now it is correctly **skipped**, not faked.
- Testing on the **real RK3588 / Orange Pi 5 Max board** is pending — no board on the bench yet.

Bottom line: boots on the Mac = proven. Real A/B slot switch + the Linux-host Android A/B test = pending / waiting on the right environment. We do not pretend anything works that has not been captured as evidence.

Page 2 — For software engineers

Fidelity ladder. T0 podman control-plane+emulator (SHIPPED, protocol-only) → T1 A/B-virt base image on QEMU virt + HVF (FOUNDATION GREEN: live-userspace boot; A/B-apply layers AUTHORED — assembler + U-Boot boot.cmd — NOT yet run, gated on u-boot.bin from the in-flight build8) → T2 Cuttlefish real Android update_engine A/B + AVB/dm-verity + auto-rollback incl. PWU-CF-2 corrupt-slot rollback (SKIP/UNVERIFIED on this host, Linux+KVM-gated) → T3 real RK3588 hardware (PENDING).

Proven (captured evidence):

- **PWU-AB-1 base image** — tests/emulator/ab_virt/out/images/Image (MD5 9f3670cd7dba7bdeeebe2c6d791e929a) + tests/emulator/ab_virt/out/images/rootfs.ext2 (MD5 a056760e88eea575977be13e38cfe430); builder tests/emulator/ab_virt/build_image.sh. (A U-Boot+RAUC rebuild is in flight and will replace these.)
- **PWU-AB-1 foundation boot** — docs/qa/20260611T061626Z-ab-virt-boot/console.log (196 lines): kernel on Apple CPU MIDR 0x610f (physical CPU ... [0x610f0000]), Buildroot buildroot login: root, post-login uname -a → Linux buildroot 6.1.44 ... aarch64, sentinel HELIX_USERSPACE_LIVE_OK (line 182), clean poweroff. §11.4.107 liveness battery (full transcript + post-login sentinel, not a single frame). Driver tests/emulator/ab_virt/boot_smoke.sh.
- **T0 round-trip** — tests/emulator/tier1_container_e2e.sh (register → update-check → telemetry asserted from captured podman logs under docs/qa/<run-id>/).
- **HelixQA bank static gate (E9)** — bash tools/helixqa/run_bank.sh --dry-run --bank tools/helixqa/banks/helix_ota.yaml → 12 passed / 0 failed / 0 skipped. HOTA-AB-VIRT-BOOT declares evidence docs/qa/20260611T061626Z-ab-virt-boot/console.log (verified present). The runner FAILs any challenge whose declared evidence is missing — this gates that the boot evidence exists + is wired into the bank (it does NOT itself exercise A/B apply).

Pending / not proven (§11.4.6):

- **A/B slot switch + dm-verity + auto-rollback (E3–E5)** = PENDING_FORENSICS. The A/B disk assembler tests/emulator/ab_virt/assemble_ab_disk.sh + U-Boot bootcount slot-select tests/emulator/ab_virt/uboot_ab/boot.cmd (bootcount / bootlimit / altbootcmd / BOOT_ORDER semantics; +uboot.env, README.md, commit 6581ab4) are AUTHORED + parse-clean but **NOT run** — gated on a real u-boot.bin from the in-flight U-Boot+RAUC build8 (commit 0d27438 added the libssl-dev/bison/flex host-tools fix after build7 failed at openssl/evp.h). No captured slot-switch or rollback evidence yet.
- **T2 Cuttlefish (E7)** = SKIP / UNVERIFIED-pending-Linux-host. Driver tests/emulator/tier2_cuttlefish_ab.sh (PWU-CF-2 corrupt-slot auto-

rollback section authored, mirrors ab_virt PWU-AB-3); host gate /dev/kvm absent on this Apple-Silicon macOS host (verified). Exit-3 honest topology SKIP per §11.4.3/§11.4.112. Script header marks the exact OTA-apply + rollback invocation UNCONFIRMED: (Virtual-A/B vs legacy A/B; update_device.py vs cvd) — runtime-detected, never guessed. Ready for the operator's incoming Linux + nested-KVM host.

- **T3 hardware (E8)** = PENDING_FORENSICS. No board.

Provenance: dd43738 (research + dev-host A/B-virt build infra) → d5374d0 (PWU-AB-1 foundation GREEN, live-userspace boot) → 4278aa9 (U-Boot + RAUC + GPT tooling) → 0f120a6 (parallel streams: HelixQA bank wiring + these Status docs + Cuttlefish PWU-CF-2 rollback) → 6581ab4 (A/B disk assembler + U-Boot bootcount slot-select, AUTHORED) → 0d27438 (build8 host-tools libssl-dev fix, build in flight). HEAD 0d27438.

§-refs: §11.4.45 (Status doc), §11.4.56 (two-audience summary), §11.4.5 (captured-evidence table), §11.4.6 (no-guessing), §11.4.107 (liveness), §11.4.3 / §11.4.112 (topology-gated SKIP), §11.4.44 (revision header).